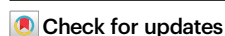


Large and tunable magnetoresistance in van der Waals ferromagnet/semiconductor junctions

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Magnetic tunnel junctions (MTJs) with conventional bulk ferromagnets separated by a nonmagnetic insulating layer are key building blocks in spintronics for magnetic sensors and memory. A radically different approach of using atomically-thin van der Waals (vdW) materials in MTJs is expected to boost their figure of merit, the tunneling magnetoresistance (TMR), while relaxing the lattice-matching requirements from the epitaxial growth and supporting high-quality integration of dissimilar materials with atomically-sharp interfaces. We report TMR up to 192% at 10 K in all-vdW Fe₃GeTe₂/GaSe/Fe₃GeTe₂ MTJs. Remarkably, instead of the usual insulating spacer, this large TMR is realized with a vdW semiconductor GaSe. Integration of semiconductors into the MTJs offers energy-band-tunability, bias dependence, magnetic proximity effects, and spin-dependent optical-selection rules. We demonstrate that not only the magnitude of the TMR is tuned by the semiconductor thickness but also the TMR sign can be reversed by varying the bias voltages, enabling modulation of highly spin-polarized carriers in vdW semiconductors.

The traditional path to enhance the TMR^{1,2} relies on carefully choosing insulators and common ferromagnets, such as MgO with Fe and Co^{3,4}. As the MTJ size scales down, this approach poses many obstacles, from materials nonuniformity and deteriorating quality to enhanced energy consumption and reduced stability⁵. The breakthroughs in vdW materials and the discovery of two-dimensional (2D) ferromagnets^{6,7} suggest important opportunities to overcome these problems in all-vdW MTJs, where realizing a large TMR ~200% could revolutionize magnetic random-access memories (MRAM)⁵.

MTJs with both conventional or vdW ferromagnets typically include an insulating spacer layer instead of the semiconducting barrier layer, owing to an extensive research on insulators such as Al₂O₃, MgO, and hBN. However, a realization of tunable spin-polarized transport in semiconductors is desirable for many emerging applications¹. Conventional materials, such as δ -doped Fe/GaAs junctions, already provide a degree of tunability with bias-dependent sign reversal of interfacial spin polarization and TMR^{8–12}. Because the observed spin-dependent signals in such systems are only modest, switching to 2D

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vdW materials could offer significant advantages by: (i) simultaneously increasing the TMR^{13–15} and supporting highly spin-polarized carriers, and (ii) expanding the tunability of spin-dependent properties, as demonstrated, for example, through barrier-thickness controlled spin polarization^{16,17} and gate-tunable magnetic proximity effects in hybridized 2D material/ferromagnet interface for spin valves^{18–23}, and gate-tunable spin galvanic effect in van der Waals heterostructures of graphene with a semimetal or topological insulator^{24,25}.

Recently, the all-vdW Fe_3GeTe_2 (FGT)-based MTJs have been widely studied, among which a large TMR of ~300% (4.2 K), ~50% (10 K), and ~110% (4.2 K) have been observed in devices with insulating spacer hBN and devices with semiconductor InSe and WSe_2 , respectively^{26–28}. Compared with an insulator, the advantage of a semiconductor tunnel barrier is that its Fermi level (E_F) can be adjusted by doping to make the E_F close to the valence band or closer to the conduction band, which plays an important role in enhancing spin-filtering effect¹³. In addition, a large room-temperature TMR of 85% was observed in $\text{Fe}_3\text{GaTe}_2/\text{WSe}_2/\text{Fe}_3\text{GaTe}_2$ MTJs²⁹, which confirms the great potential for semiconductor-based MTJs. 2D gallium selenide (GaSe) crystal is a typical layered metal monochalcogenide with an indirect bandgap energy of ~2 eV in the bulk³⁰, which can serve as a perfect tunnel barrier³¹. Furthermore, it was predicted that the giant magnetoresistance can be obtained by using the semiconductor barrier due to the spin-filtering effect^{13,15}. However, the magnetoresistance properties of all-vdW MTJs with GaSe barriers have not been reported yet.

In this work, we not only find that the magnitude of TMR increases first and then decreases with increasing the thickness of the semiconductor spacer GaSe in $\text{Fe}_3\text{GeTe}_2/\text{GaSe}/\text{Fe}_3\text{GeTe}_2$ MTJs, but also find the magnitude and sign of the TMR can be tuned by the bias voltage. The maximum TMR of up to 192% is obtained with 9 layers of a GaSe spacer. This realization greatly expands materials design opportunities

for semiconductor spintronics that are unavailable to MTJs with insulators^{32,33}, including applications in artificial neural networks^{34,35} and spin lasers³⁶.

Results

The typical optical image of the core structure of the MTJ devices is plotted in Fig. 1a, where the two FGT electrodes are separated by a GaSe layer. To avoid oxidation, we covered the core structure of the device with a hBN flake. The schematic diagram of the device and magnetotransport setup is shown in the inset of Fig. 1a, where an out-of-plane magnetic field $\mathbf{B}$ controls the magnetization alignment of the FGT electrodes. Our previous works confirmed that GaSe and FGT have high-quality crystal structure^{28,37–40}. The photoluminescence spectrum measurement shows that GaSe has a bandgap of ~2 eV (Supplementary Fig. 1). The MTJ devices (A, B, C, D, E, F, and G) with different GaSe-layer thicknesses were fabricated using mechanical exfoliation and dry transfer method (see “Methods”), where the GaSe-layer thicknesses were determined by atomic force microscope (AFM) for device A, B, C, D, E, F, and G are about 5.5, 6.5, 7.3, 8.2, 9.2, 10.0, and 15.6 nm, respectively (Supplementary Fig. 2). To have different coercive fields for the top and bottom electrodes, we select different thicknesses of FGT flakes^{28,39,40}. The typical thickness of the bottom and top FGT is ~8 nm and ~12 nm, respectively. From the optical image of all the fabricated devices, the active junction overlap areas $A < 20 \mu\text{m}^2$, which are comparable to the typical magnetic domain sizes in FGT flakes⁴¹.

We first investigate the current-voltage (I - V_{bias}) characteristics under applied perpendicular $B = -0.4$ T to ensure the parallel-magnetization configuration of the two FGT. To directly compare different devices, the normalized nonlinear current density-voltage J - V_{bias} curves at 10 K for devices A to G are shown in Fig. 1b, where the nonlinear behavior of devices A and B with a thinner GaSe layer are

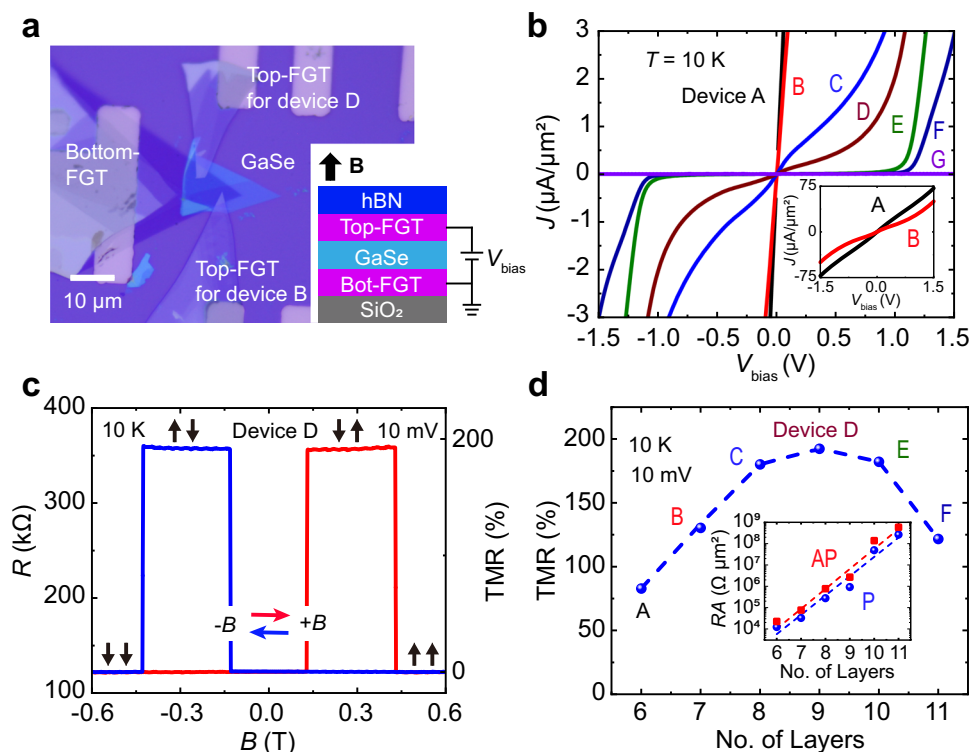


Fig. 1 | Large TMR in the FGT/GaSe/FGT MTJ devices. **a** Typical optical image of the core structure of the device made of different flakes. Inset shows the schematic diagram of the device and magnetotransport setup. The magnetic field ($\mathbf{B}$) is applied in an out-of-plane direction. **b** Current density/versus applied bias V_{bias} for the devices with the GaSe thickness ranging from 5.5 to 15.6 nm (devices A–G) in parallel-magnetic configuration. The inset shows the J - V_{bias} curves of devices A and

B in a larger bias range. **c** Magnetic hysteresis of the resistance R loop for device D at $V_{\text{bias}} = 10$ mV, and the corresponding TMR is ~192.4%. Red and blue horizontal arrows show the sweeping directions of $\mathbf{B}$. Black-vertical arrows denote the two FGTs' magnetization configurations. **d** The measured maximum TMR ratios in the different devices at 10 mV. The inset shows the plotted zero-bias log(RA) is nearly linear with the number of GaSe layers in both parallel and antiparallel states.

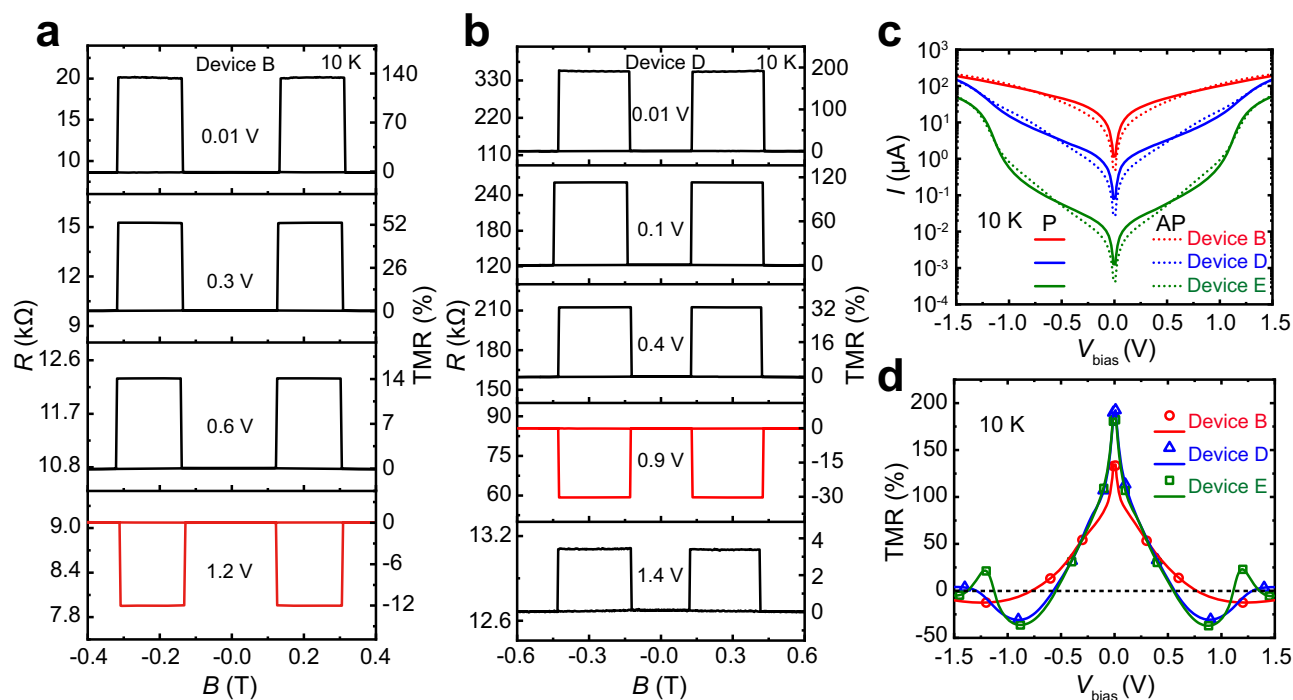


Fig. 2 | The bias-dependent TMR of the devices. a, b The R - B curves at various positive bias for devices B and D. **c** I - V_{bias} curves of devices B, D, and E in parallel and

antiparallel states, respectively. **d** The corresponding TMR as a function of V_{bias} . The hollow symbols are extracted from the R - B curves. The temperature is fixed at 10 K.

shown in a larger current range (inset of Fig. 1b). The nonlinear J - V_{bias} characteristics as well as the obvious parabolic R - V_{bias} curves (Supplementary Fig. 3) reveal the existence of the tunnel barrier between GaSe and FGTs. The band alignment at the FGT/GaSe interface, obtained by analyzing the transport mechanism^{42,43}, also reveals that the effective tunnel barrier of electrons is up to -0.9 eV (Supplementary Fig. 4). In some previous studies, the effective tunnel-barrier heights of CrI_3/hBN , Fe/MgO and CoFe/MgO interfaces are estimated to be 0.25 eV⁴⁴, 0.39 eV⁴, and 0.9 eV⁴⁵, respectively. Therefore, the tunnel-barrier height of 0.9 eV in the FGT/GaSe interface could ensure the tunneling transport of electrons.

We next examine the TMR. Upon sweeping the out-of-plane **B**, the devices A-F show two distinct parallel (R_{P}) and antiparallel resistance (R_{AP}) states (Supplementary Fig. 5). Among them, at bias of 10 mV, the R_{P} and R_{AP} of device D are 122.25 k Ω and 357.52 k Ω , respectively (Fig. 1c). The corresponding $\text{TMR} = (R_{\text{AP}} - R_{\text{P}})/R_{\text{P}}$ is 192.4% . The thickness of GaSe-layer dependence of the measured maximum TMR ratio is shown in Fig. 1d, which first increases from 83.1% (device A) and 133.5% (device B) and 180.2% (device C) to 192.4% (device D) and then decreases to 182.3% (device E) and 121.7% (device F), and finally vanishes (devices G) with increasing the thickness of GaSe layer. The zero-bias resistance-area product (RA) of the devices for both the parallel and antiparallel states increases approximately exponentially with increasing the GaSe thickness, suggesting that the transport mechanism is dominated by tunneling (inset of Fig. 1d)⁴⁶. The large TMR and nonlinear J - V curve indicate that the GaSe spacer serves as a good tunnel barrier. The maximum TMR at low bias is found in the device with GaSe of 8.2 nm, and the internal physical mechanism can be explained as follows. On the one hand, the spin-filtering effect of GaSe gets weaker when the thickness is reduced, resulting in the decrease of TMR^{2,13}. On the other hand, with further increase of the spacer thickness, the TMR decreases and, eventually, vanishes in device G with 15.6 -nm-thick GaSe, where the tunneling through extrinsic defects could become important and eventually exceed the spin-relaxation length of GaSe^{47,48}. To confirm the repeatability of this barrier-thickness-dependent TMR behavior,

we fabricated and measured another group of devices with variable barrier thickness (Supplementary Fig. 6). Similar layer-thickness-dependent TMR was proved, further confirming that the magnitude of TMR is directly related to the semiconductor layer thickness, which can be improved by optimizing the thickness and quality of the barrier layer.

To investigate $\text{TMR}(V_{\text{bias}})$ for devices with different GaSe thickness, we measured the R - B curves. As shown in Fig. 2a, b, the positive TMRs decrease with V_{bias} . Negative TMRs of -12.3% and -30.5% for devices B and D are obtained at 1.2 V and 0.9 V, respectively. This salient sign inversion of TMR is found in all the devices A-F. The number of sign reversals of TMR can be tuned, from single to multiple, with increasing the GaSe thickness. To better understand the variation of TMR with bias, as shown in Fig. 2c, we measured the I - V_{bias} curves of the devices in parallel and antiparallel states, respectively. The nonlinear I - V_{bias} curves for devices B, D, and E show very different trends in parallel and antiparallel states, which allow us to derive bias-dependent TMR for these devices (Fig. 2d). The obtained TMR value matches well to that extracted from the R - B curves (Fig. 2a, b and Supplementary Figs. 7 and 8), indicating the influence of the Zeeman effect on TMR is negligible. The symmetric bias-dependent current and TMR suggest the symmetrical FGT/GaSe interfaces in these devices.

Devices A (Supplementary Fig. 9) and B show similar bias-dependent behavior of TMR. Specifically, as shown in Fig. 2c, for device B, the measured current for the parallel state is higher than that for the antiparallel state for $V_{\text{bias}} < 0.76$ V, leading to a positive TMR. However, beyond such V_{bias} , the measured current for the parallel state is lower than that for the antiparallel state, resulting in a negative TMR (Fig. 2d). With increasing the thickness of the GaSe spacer layer, we observed multiple sign changes of the TMR. As shown in Fig. 2c, the nonlinear I - V_{bias} curves of device D in parallel and antiparallel states show two crossovers, and similar behavior is also observed in device C (Supplementary Fig. 10). Correspondingly, in Fig. 2d, the TMR of device D first decreases monotonically and changes sign of around 0.58 V, and again around 1.27 V as the bias increases. With further increasing the GaSe thickness, as shown in Fig. 2c, the nonlinear I - V_{bias}

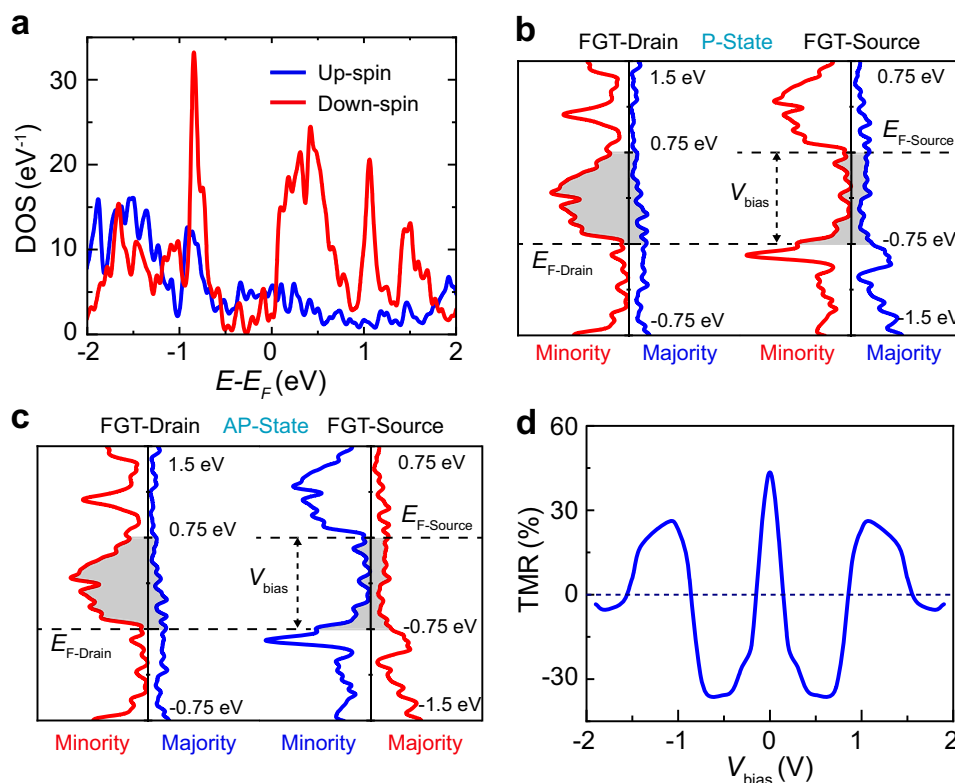


Fig. 3 | The simulation of spin-resolved DOS of FGT and bias-dependent TMR calculated by the elastic tunneling model. a The calculated spin-resolved DOS for both the up (blue line) and down (red) spins of the FGT in 3-layer-FGT/6-layer-GaSe/3-layer-FGT heterojunction. **b, c** Schematic diagram of

direct band-to-band spin-dependent tunneling under bias window V_{bias} (shaded area) in the parallel and antiparallel states, respectively. **d** The calculated TMR as a function of V_{bias} by using the simple elastic tunneling formula.

curves reveal three crossovers for device E. In Fig. 2d, the TMR of device E first decreases monotonically and changes sign around 0.56 V, and then the TMR decreases in an oscillatory fashion as the bias increases. Similar three-sign changes of TMR behavior is also observed in device F (Supplementary Fig. 11).

To understand the bias-dependent magnetotransport, the spin-resolved density of states (DOS) of FGT in the 3-layer-FGT/6-layer-GaSe/3-layer-FGT heterojunction were obtained using the first-principles calculations (see Supplementary Fig. 12). The calculated spin-resolved DOS of FGT electrode is shown in Fig. 3a. Assuming that tunneling is elastic and spin-conserving, with the corresponding tunneling probability independent of the initial and final states as well as the tunnel-barrier height for electrons⁴⁹, the spin-dependent tunneling current at zero temperature can be expressed as^{1,2,50}

$$I_{\sigma} \propto \int_{\mu_D}^{\mu_S} \rho_D^{\sigma}(E - \mu_D) \rho_S^{\sigma}(E - \mu_S) dE, \quad (1)$$

where the $\mu_{D(S)}$ and $\rho_{D(S)}^{\sigma}$ are the chemical potentials and spin-resolved DOS for the drain (or source) FGT, respectively (σ is the spin index), and $\mu_D - \mu_S = eV_{\text{bias}}$. The tunneling currents in parallel (Fig. 3b) and antiparallel (Fig. 3c) configurations can thus be expressed as

$$I_P \propto \int_{\mu_D}^{\mu_S} (\rho_D^{\uparrow}(E - \mu_D) \rho_S^{\uparrow}(E - \mu_S) + \rho_D^{\downarrow}(E - \mu_D) \rho_S^{\downarrow}(E - \mu_S)) dE, \quad (2)$$

and

$$I_{AP} \propto \int_{\mu_D}^{\mu_S} (\rho_D^{\uparrow}(E - \mu_D) \rho_S^{\downarrow}(E - \mu_S) + \rho_D^{\downarrow}(E - \mu_D) \rho_S^{\uparrow}(E - \mu_S)) dE. \quad (3)$$

The resulting bias-dependent TMR is shown in Fig. 3d. With increasing bias, the calculated TMR rapidly drops, changes sign, and then oscillates in qualitative agreement with the measurements for devices E and F. Multiple sign changes of TMR with increasing bias have also been predicted theoretically for Fe/MoS₂/Fe MTJ devices¹³. In addition, the multiple sign changes of TMR have also been observed in FGT/hBN/FGT, FGT/WSe₂/FGT, and FGT/WS₂/FGT MTJs²⁷. In our FGT/GaSe/FGT MTJs, within the range of 1.5 V bias, as the thickness of a GaSe spacer decreases, the number of TMR reversals decreases to two in devices C and D and to only one in devices A and B. A possible physical mechanism of the thickness-dependent TMR reversals is explained as follows.

The above calculation of the spin-dependent tunneling current assumed that the transmission probability is the same for all initial and final states. However, if tunneling is at least partially coherent, the transmission probability should be larger for states with transverse momenta close to those where the decay rate of the evanescent states in GaSe is the smallest. Because the bandgap in GaSe is indirect, this may occur away from the Gamma point⁵¹. The efficiency of this transverse-momentum filtering increases with increasing thickness of the tunnel barrier⁵⁰.

Because the interlayer dispersion of the vdW FGT states is weak, they are at a given transverse momentum essentially quantized. An energy isosurface of a lead thus consists of one or more 1D Fermi contours. Coherent tunneling is only possible at the intersections of the isosurfaces of the two leads corresponding to the same electrochemical potential. At large barrier thickness, a strong enhancement of the tunneling current should occur when such crossing points fall close to the points of the lowest decay rate in GaSe. If such matching condition is satisfied at the given bias for initial and final states of the

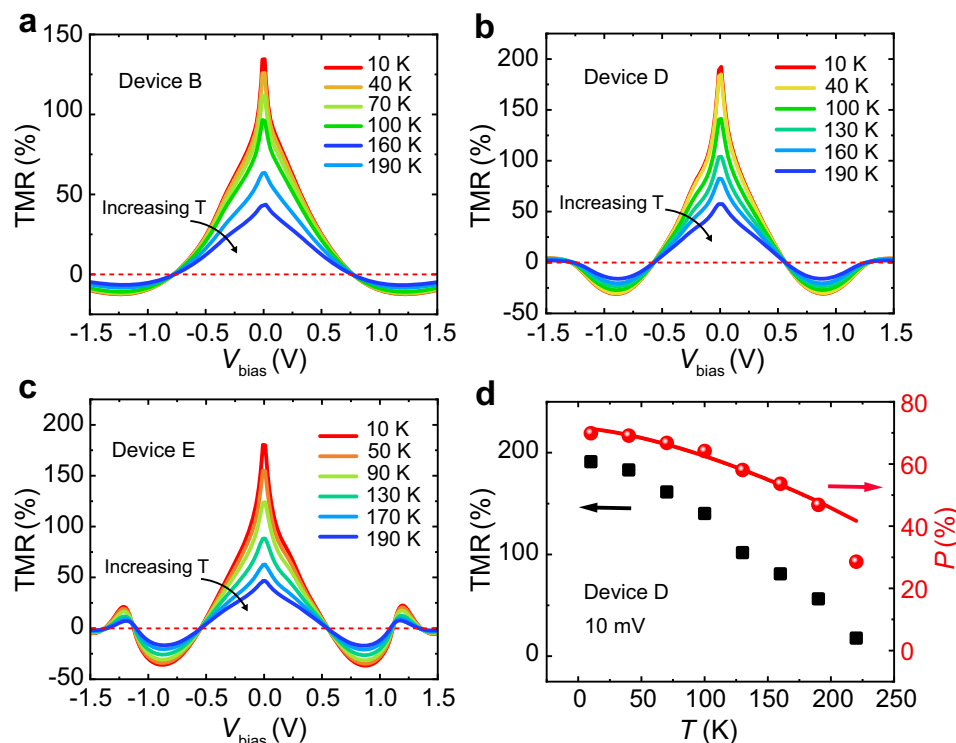


Fig. 4 | The temperature-dependent TMR. a–c TMR ratios of devices B, D, and E measured at temperatures from 10 to 190 K, respectively. **d** The TMR and spin

polarization of device D as a function of temperature at bias of 10 mV. The red line shows the fitting data by the Bloch's law¹.

same or opposite spin, positive or negative TMR is expected, respectively (Supplementary Fig. 13). The relative importance of such matching should increase at a larger barrier thickness, as long as the coherent tunneling persists. Thus, coherent tunneling may explain why additional sign changes of TMR as a function of bias are observed at larger GaSe thicknesses.

On the other hand, the barriers thickness-dependent TMR reversals can be related to the tunnel junction resistance values. With semiconducting tunnel barriers as thick as the GaSe film in device E, the elastic tunneling model with FGT electronic structures can be applicable since all the voltage drops of the device should be happening at the tunnel junctions. With thinner semiconducting films and lower tunnel junction resistances, however, the bias voltage effectively applied to the tunnel junctions should be smaller than the voltage driven by an external voltage source: this phenomenon is often dubbed as a voltage-divider effect. To prove this, we extracted the bias voltages where the TMR-sign reversals firstly occur for different devices (Supplementary Fig. 14), the bias voltages increases as the tunnel junction resistance decreases, indicating the voltage-divider effect could also be the reason for the barriers thickness-dependent TMR reversals.

We further investigate the temperature-dependent TMR effect in our devices. The I - V_{bias} curves of devices B, D, and E in parallel and antiparallel states were measured at temperatures from 10 K to 190 K. The corresponding TMR is plotted in Fig. 4a–c for device B, D, and E, respectively. The TMR- V_{bias} curves in Fig. 4a–c all pass through the same zero at all temperatures, verifying that the bias-dependent TMR in the devices is dominated by tunneling⁴². The extracted TMR at different temperatures at 10 mV for device D (Fig. 4d) and device B and E (Supplementary Fig. 15) show a decreasing trend, which can be attributed to the decrease of spin polarization with temperature. When V_{bias} is extremely small, approximately only the electrons at the E_F take part in tunneling transport. For simplicity, assume the source and drain FGT electrodes have almost the same

spin polarization. Then TMR can be defined as $\text{TMR} = 2P^2/(1 - P^2)$, where P denote the spin polarization at the E_F for the drain and source FGT electrode^{1,2}. As shown in Fig. 4d, the P decreases with temperature, and the maximum P at 10 K is up to 70%, which is larger than that obtained in other 2D semiconductor-based MTJs, such as 45% in $\text{Fe}_3\text{GeTe}_2/\text{InSe}$ and 60% in $\text{Fe}_3\text{GeTe}_2/\text{WSe}_2$ interfaces^{27,28}, but lower than that of $\text{Ni(111)}/\text{Gr}$ interface⁵². The estimated temperature-dependence of the spin polarization can be fitted well by Bloch's law, given by $P = P_0(1 - \alpha T^{3/2})$, where P_0 is the spin polarization at 0 K, α is a materials-dependent constant⁵³. The fitting value of α is $1.26 - 1.40 \times 10^{-4} \text{ K}^{-3/2}$, which is comparable to the previous reports^{26,40}.

Our presented results, which demonstrate large and tunable TMR, substantiate an ambitious vision where all-vdW MTJs could replace various charge-based memory applications⁵, targeted to reach TMR ~200% for commercial viability. Implementing such vdW MTJs is expected to rely on insulating 2D tunnel barrier⁵, just as it was shown with hBN barrier in an all-vdW MTJ with the low-temperature TMR ~160%²⁶ and also supported by a very recent report of TMR ~300%²⁷. However, since we observe the desired large TMR values even with a semiconductor spacer, the prospect for all-vdW spintronics becomes considerably broader than just memory applications and the resulting large spin polarization and spin-orbit coupling opens opportunities beyond magnetoresistive effects. For example, the measured sign reversal of the TMR with applied bias is consistent with the reversal of the carrier spin polarization and could enable desirable polarization modulation³⁶. Our findings could integrate semiconductor-based optoelectronics, microelectronics, and spintronics together, and be also relevant to emerging cryogenic applications where proximity-modified semiconductors and MTJs provide a platform for fault-tolerant quantum computing⁵⁴. Given the continued advances in understanding of vdW semiconductor/ferromagnet junctions, including a surprising optical manifestation of the valley-dependent magnetic proximity effect^{55,56}, we expect further opportunities for valley-

dependent transport properties by using semiconducting tunnel barriers in MTJs.

Methods

Fabrication of the $\text{Fe}_3\text{GeTe}_2/\text{GaSe}/\text{Fe}_3\text{GeTe}_2$ MTJ devices. The high-quality vdW bulk single-crystal FGT and hBN were purchased from HQ Graphene, while GaSe was purchased from 2D semiconductors, respectively. Firstly, a FGT flake was exfoliated onto polydimethylsiloxane (PDMS) stamps by adhesive tape. The stamps were adhered to a glass slide. Under optical microscope, the FGT flake with appropriate thickness and shape was chosen to transfer onto a 300-nm-thick SiO_2/Si substrate by using a position-controllable dry transfer method⁴⁰. Then, using the same method, a GaSe flake was transferred onto the FGT flake, followed by another thicker FGT flake to fabricate a 2D heterojunction. To prevent the FGT from oxidation, a 20 nm-thick hBN layer was used to cap the whole heterostructure stack. Finally, the device was annealed at 120 °C for 10 min to remove the bubbles between the layers and ensure close contact between the layers. Notably, the whole transfer processes were performed in a nitrogen-filled glovebox with a concentration of less than 1 ppm of oxygen and water to ensure a clean interface. The source and drain electrode regions were pre-patterned by standard photolithography, and Cr/Au (10/40 nm) layers were deposited using an ultrahigh vacuum magnetron sputtering system, followed by a lift-off process. The thicknesses of both GaSe and FGT flakes were measured by using AFM (Bruker Multimode 8).

Measurements of MR effect. Electrical properties were measured using a semiconductor characterization system (Agilent Technology B1500A). All measurements were carried out in a Model CRX-VF Cryogenic Probe Station with a ± 2.5 T out-of-plane vertical magnetic field.

Ab Initio Calculations. Our first-principles calculations are performed by the density functional theory (DFT) using the Vienna ab initio Simulation Package (VASP) code. The details are shown in the Supplementary Note 7.

Data availability

The data that support the findings of this study are available within the article and the Supplementary Information or available from the corresponding author upon reasonable request. All data generated in this study are provided in the Supplementary Information/Source Data file. Source data are provided with this paper.

References

- Žutić, I., Fabian, J. & Das Sarma, S. Spintronics: fundamentals and applications. *Rev. Mod. Phys.* **76**, 323–410 (2004).
- Miao, G.-X., Münzenberg, M. & Moodera, J. S. Tunneling path toward spintronics. *Rep. Prog. Phys.* **74**, 036501 (2011).
- Parkin, S. S. P. et al. Giant tunnelling magnetoresistance at room temperature with MgO (100) tunnel barriers. *Nat. Mater.* **3**, 862–867 (2004).
- Yuasa, S. et al. Giant room-temperature magnetoresistance in single-crystal Fe/MgO/Fe magnetic tunnel junctions. *Nat. Mater.* **3**, 868–871 (2004).
- Yang, H. et al. Two-dimensional materials prospects for non-volatile spintronic memories. *Nature* **606**, 663–673 (2022).
- Gong, C. et al. Discovery of intrinsic ferromagnetism in two-dimensional van der Waals crystals. *Nature* **546**, 265–269 (2017).
- Huang, B. et al. Layer-dependent ferromagnetism in a van der Waals crystal down to the monolayer limit. *Nature* **546**, 270–273 (2017).
- Chantis, A. N. et al. Reversal of spin polarization in Fe/GaAs (001) driven by resonant surface states: first-principles calculations. *Phys. Rev. Lett.* **99**, 196603 (2007).
- Fujita, Y. et al. Nonmonotonic bias dependence of local spin accumulation signals in ferromagnet/semiconductor lateral spin-valve devices. *Phys. Rev. B* **100**, 024431 (2019).
- Crooker, S. A. et al. Imaging spin transport in lateral ferromagnet/semiconductor structures. *Science* **309**, 2191–2195 (2005).
- Lou, X. et al. Electrical detection of spin transport in lateral ferromagnet–semiconductor devices. *Nat. Phys.* **3**, 197–202 (2007).
- Moser, J. et al. Bias dependent inversion of tunneling magnetoresistance in Fe/GaAs/Fe tunnel junctions. *Appl. Phys. Lett.* **89**, 162106 (2006).
- Dolui, K., Narayan, A., Rungger, I. & Sanvito, S. Efficient spin injection and giant magnetoresistance in Fe/MoS₂/Fe junctions. *Phys. Rev. B* **90**, 041401(R) (2014).
- Li, X. et al. Spin-dependent transport in van der Waals magnetic tunnel junctions with Fe₃GeTe₂ electrodes. *Nano Lett.* **19**, 5133–5139 (2019).
- Zhang, L. et al. Perfect spin filtering effect on Fe₃GeTe₂-based Van der Waals magnetic tunnel junctions. *J. Phys. Chem. C* **124**, 27429–27435 (2020).
- Kamalakar, M. V., Dankert, A., Kelly, P. J. & Dash, S. P. Inversion of spin signal and spin filtering in ferromagnet/hexagonal boron nitride-graphene van der Waals heterostructures. *Sci. Rep.* **6**, 21168 (2016).
- Piquemal-Banci, M. et al. Insulator-to-metallic spin-filtering in 2D-magnetic tunnel junctions based on hexagonal boron nitride. *ACS Nano* **12**, 4712–4718 (2018).
- Xu, J. et al. Spin inversion in graphene spin valves by gate-tunable magnetic proximity effect at one-dimensional contacts. *Nat. Commun.* **9**, 2869 (2018).
- Lazić, P., Belashchenko, K. D. & Žutić, I. Effective gating and tunable magnetic proximity effects in two-dimensional heterostructures. *Phys. Rev. B* **93**, 241401(R) (2016).
- Lazić, P., Sipahi, G. M., Kawakami, R. K. & Žutić, I. Graphene spintronics: Spin injection and proximity effects from first principles. *Phys. Rev. B* **90**, 085429 (2014).
- Leutenantsmeyer, J. C., Kaverzin, A. A., Wojtaszek, M. & van Wees, B. J. Proximity induced room temperature ferromagnetism in graphene probed with spin currents. *2D Mater.* **4**, 014001 (2017).
- Karpiak, B. et al. Magnetic proximity in a van der Waals heterostructure of magnetic insulator and graphene. *2D Mater.* **7**, 015026 (2020).
- Piquemal-Banci, M. et al. Spin filtering by proximity effects at hybridized interfaces in spin-valves with 2D graphene barriers. *Nat. Commun.* **11**, 5670 (2020).
- Hoque, A. M. et al. All-electrical creation and control of spin-galvanic signal in graphene and molybdenum ditelluride heterostructures at room temperature. *Commun. Phys.* **4**, 124 (2021).
- Khokhriakov, D., Hoque, A. M., Karpiak, B. & Dash, S. P. Gate-tunable spin-galvanic effect in graphene-topological insulator van der Waals heterostructures at room temperature. *Nat. Commun.* **11**, 3657 (2020).
- Wang, Z. et al. Tunneling spin valves based on Fe₃GeTe₂/hBN/Fe₃GeTe₂ van der Waals heterostructures. *Nano Lett.* **18**, 4303–4308 (2018).
- Min, K.-H. et al. Tunable spin injection and detection across a van der Waals interface. *Nat. Mater.* **21**, 1144–1149 (2022).
- Zhu, W. et al. Large tunneling magnetoresistance in van der Waals ferromagnet/semiconductor heterojunctions. *Adv. Mater.* **33**, 2104658 (2021).
- Zhu, W. et al. Large room-temperature magnetoresistance in van der Waals ferromagnet/semiconductor junctions. *Chin. Phys. Lett.* **39**, 128501 (2022).
- Rahaman, M. et al. Vibrational properties of GaSe: a layer dependent study from experiments to theory. *Semicond. Sci. Tech.* **33**, 125008 (2018).
- Kurtin, S., McGill, T. C. & Mead, C. A. Tunneling currents and the E-k relation. *Phys. Rev. Lett.* **25**, 756 (1970).
- Žutić, I. et al. Proximitized materials. *Mater. Today* **22**, 85–107 (2019).

33. Sierra, J. F. et al. Van der Waals heterostructures for spintronics and opto-spintronics. *Nat. Nanotechnol.* **16**, 856–868 (2021).
34. Sangwan, V. K. & Hersam, M. C. Neuromorphic nanoelectronic materials. *Nat. Nanotechnol.* **15**, 517–528 (2020).
35. Grollier, J. et al. Neuromorphic spintronics. *Nat. Electron.* **3**, 360–370 (2020).
36. Lindemann, M. et al. Ultrafast spin-lasers. *Nature* **568**, 212–215 (2019).
37. Cao, Y. et al. Strong enhancement of photoresponsivity with shrinking the electrodes spacing in few layer GaSe photodetectors. *Sci. Rep.* **5**, 8130 (2015).
38. Yan, F. et al. Fast, multicolor photodetection with graphene-contacted p-GaSe/n-InSe van der Waals heterostructures. *Nanotechnology* **28**, 27LT01 (2017).
39. Hu, C. et al. From two- to multi-state vertical spin valves without spacer layer based on Fe₃GeTe₂/van der Waals homo-junctions. *Sci. Bull.* **65**, 1072–1077 (2020).
40. Lin, H. et al. Spin-valve effect in Fe₃GeTe₂/MoS₂/Fe₃GeTe₂ van der Waals heterostructures. *ACS Appl. Mater. Interfaces* **12**, 43921–43926 (2020).
41. Fei, Z. et al. Two-dimensional itinerant ferromagnetism in atomically thin Fe₃GeTe₂. *Nat. Mater.* **17**, 778–782 (2018).
42. Chiu, F.-C. A review on conduction mechanisms in dielectric films. *Adv. Mater. Sci. Eng.* **2014**, 578168 (2014).
43. Simmons, J. G. Generalized formula for the electric tunnel effect between similar electrodes separated by a thin insulating film. *J. Appl. Phys.* **34**, 1793–1803 (1963).
44. Wang, Z. et al. Very large tunneling magnetoresistance in layered magnetic semiconductor CrI₃. *Nat. Commun.* **9**, 2516 (2018).
45. Moodera, J. S. & Kinder, L. R. Ferromagnetic–insulator–ferromagnetic tunneling: spin-dependent tunneling and large magnetoresistance in trilayer junctions. *J. Appl. Phys.* **79**, 4724–4729 (1996).
46. Tsunekawa, K. et al. Giant tunneling magnetoresistance effect in low-resistance CoFeB/MgO(001)/CoFeB magnetic tunnel junctions for read-head applications. *Appl. Phys. Lett.* **87**, 072503 (2005).
47. Jiang, X., Panchula, A. F. & Parkin, S. S. P. Magnetic tunnel junctions with ZnSe barriers. *Appl. Phys. Lett.* **83**, 5244–5246 (2003).
48. Takasuna, S. et al. Weak antilocalization induced by Rashba spin-orbit interaction in layered III–VI compound semiconductor GaSe thin films. *Phys. Rev. B* **96**, 161303(R) (2017).
49. Meservey, R. & Tedrow, P. M. Spin-polarized electron tunneling. *Phys. Rep.* **238**, 173–243 (1994).
50. Tsybal, E. Y. & Žutić, I. *Handbook of Spin Transport and Magnetism* Ch. 13 (CRC Press, (2019)).
51. Zatzko, V. et al. Band-structure spin-filtering in vertical spin valves based on chemical vapor deposited WS₂. *ACS Nano* **13**, 14468–14476 (2019).
52. Zatzko, V. et al. Almost perfect spin filtering in graphene-based magnetic tunnel junctions. *ACS Nano* **16**, 14007–14016 (2022).
53. Evans, R. F. L., Atxitia, U. & Chantrell, R. W. Quantitative simulation of temperature-dependent magnetization dynamics and equilibrium properties of elemental ferromagnets. *Phys. Rev. B* **91**, 144425 (2015).
54. Zhou, T. et al. Tunable magnetic textures in spin valves: from spintronics to Majorana bound states. *Phys. Rev. B* **99**, 134505 (2019).
55. Zhou, T. & Žutić, I. Asymmetry in the magnetic neighbourhood. *Nat. Mater.* **22**, 284–285 (2023).
56. Choi, J., Lane, C., Zhu, J.-X. & Crooker, S. A. Asymmetric magnetic proximity interactions in MoSe₂/CrBr₃ van der Waals heterostructures. *Nat. Mater.* **22**, 305–310 (2023).

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Author contributions

K.W. conceived the work. W.Z. fabricated the devices, W.Z., F.Y., Y.D., H.L., and Z.W. performed the experiments. W.Z., K.B., and K.W. analyzed the data. Y.Z., T.Z., X.Z., Q.C., H.Y., I.Ž., K.B., and K.W. carried out the modeling. W.Z., H.Y., L.Z., I.Ž., K.B., and K.W. wrote the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Supplementary Information

Large and tunable magnetoresistance in van der Waals Ferromagnet/Semiconductor junctions

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References for Supplementary Information

Supplementary Note 1. Characterization of phase structure and conductivity of GaSe

Gallium selenide (GaSe) single crystals are a typical two-dimensional (2D) layered metal monochalcogenide with an indirect bandgap energy of ~ 2 eV in the bulk^{1,2}. To characterize the crystal quality of GaSe, we measured the Raman and PL spectrum of the bulk GaSe. As shown in **Figure S1a**, the Raman spectrum of the bulk GaSe shows peaks at ~ 135 cm^{-1} , 214 cm^{-1} , 245 cm^{-1} , and 308 cm^{-1} , corresponding to the A_{1g}^2 , E_{2g}^1 , E_{1g}^2 , and A_{1g}^2 vibration mode of GaSe, respectively. The PL measurements indicates the bandgap energy of the bulk GaSe is ~ 2 eV (**Figure S1b**). This notation is consistent with earlier works on bulk ϵ -GaSe^{1,3}.

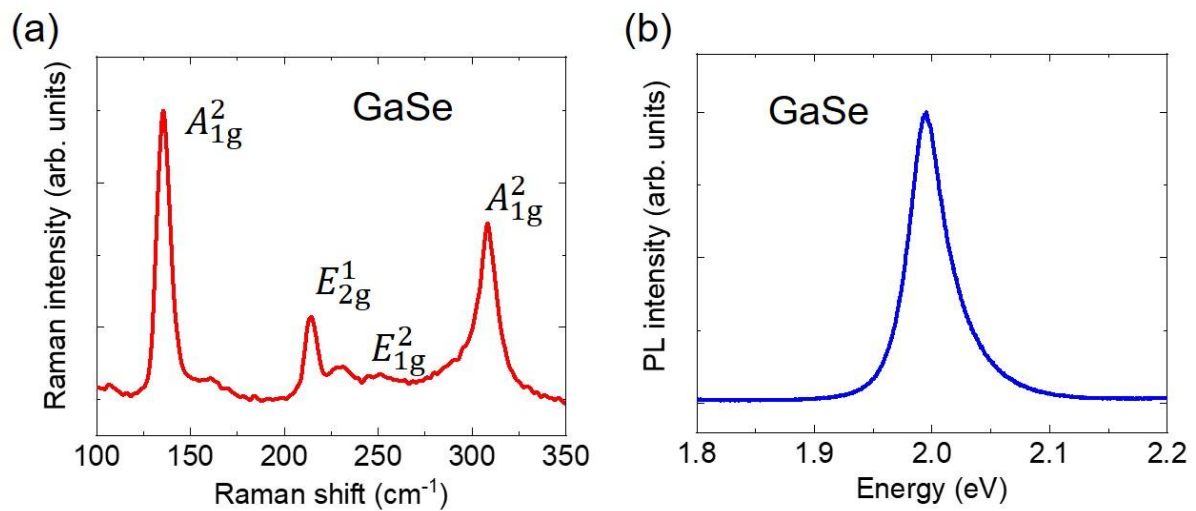


Figure S1. (a) The Raman spectrum and (b) Photoluminescence (PL) spectrum of a bulk GaSe flake (measured with 532 nm laser excitation at room temperature). The spectrum shows the band edge emission of GaSe, which is centered at about 2 eV.

Supplementary Note 2. The thickness of the GaSe barrier layers

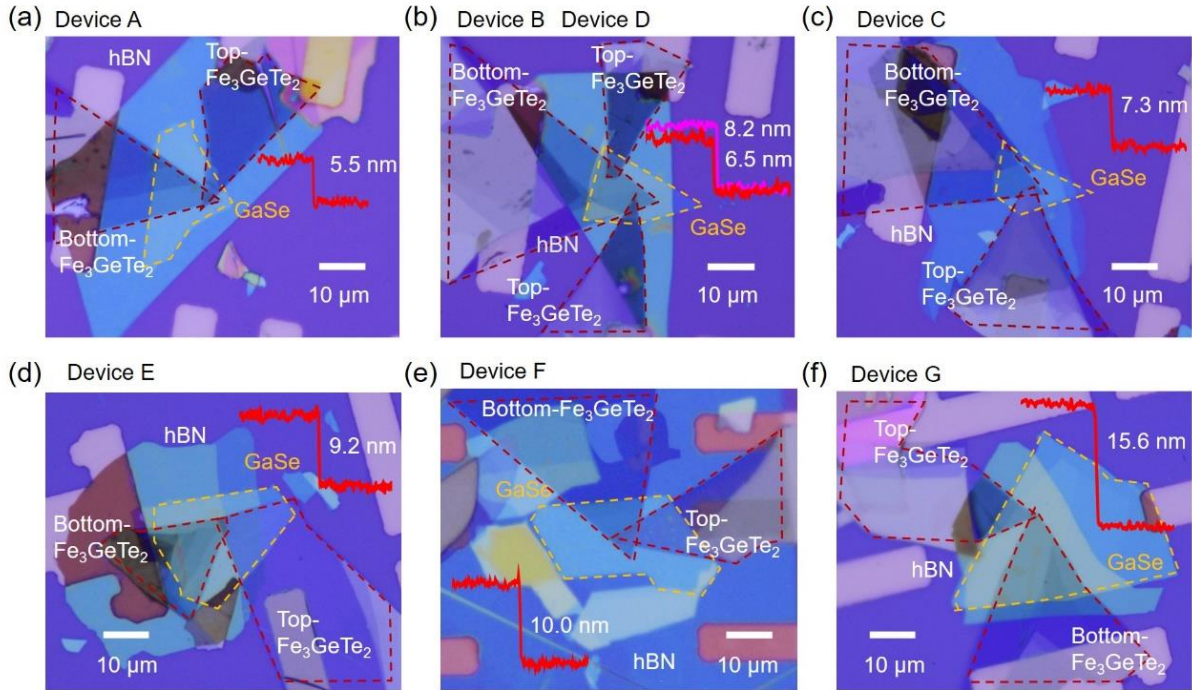


Figure S2. (a-f) The optical images of the devices A-G, respectively. The insets of the figures (a-f) show the scanning thickness of GaSe flakes for devices A-G by AFM measurements. The FGT and GaSe flakes are outlined with red and orange dotted lines respectively. The scale bar is 10 μm.

As shown in **Figure S2a-f**, the optical images of the devices A, B, C, D, E, F and G, where The FGT and GaSe flakes are outlined with red and orange dotted lines respectively. The insets show the scanned thickness of the GaSe flakes by atomic force microscope (AFM), indicating the devices A, B, C, D, E, F and G is about 5.5, 6.5, 7.3, 8.2, 9.2, 10.0 and 15.6 nm, respectively. Due to the thickness of monolayer of GaSe is about 0.93 nm⁴, thus the devices A, B, C, D, E, F and G have 6-layers, 7-layers, 8-layers, 9-layers, 10-layers, 11-layers and 17-layers of GaSe monolayer, respectively.

Supplementary Note 3. Tunneling mechanism of the MTJ devices

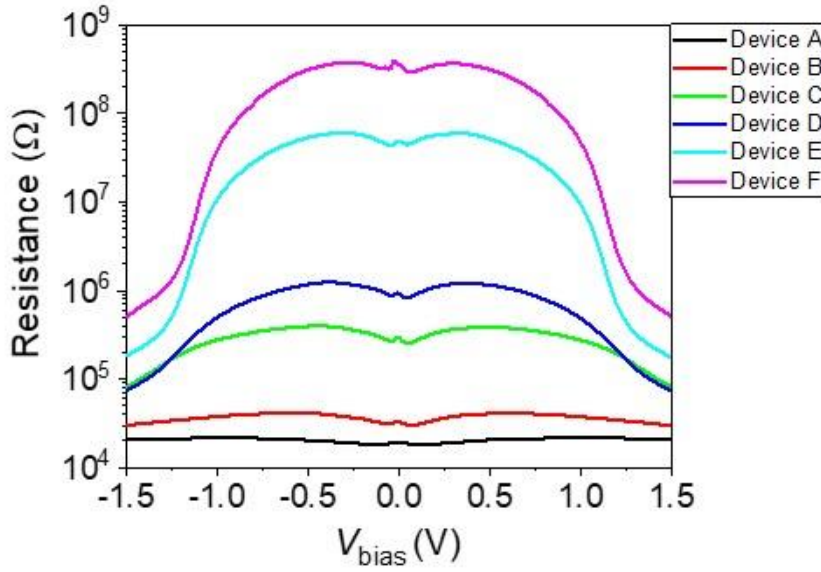


Figure S3. The resistances of the devices change with the bias voltage. In different devices, the change trend of the resistance with the bias voltage is almost the same. Under a large bias voltage, the resistances of the devices show parabolic behavior, indicating that they are all tunneling devices.

We next analyze the tunneling mechanisms in the different devices. When the applied bias voltage is far less than the effective tunnel-barrier height ($V_{\text{bias}} \ll \phi_B$), the conduction mechanism is only direct tunneling, and current density-voltage dependence of direct tunneling is described by the Simmons' formula⁵, that is,

$$J = \frac{3q^2(2m^*q\phi_B)^{1/2}V_{\text{bias}}}{2h^2d} \exp\left[-\frac{4\pi d(2m^*q\phi_B)^{1/2}}{h}\right], \quad (1)$$

where q is the elementary charge, d is the barrier width, m^* is the effective electron mass ($\sim 0.1 m_0$ in GaSe⁶), ϕ_B is the effective tunnel-barrier height and h is Planck's constant. When the applied bias voltage exceeds the effective tunnel-barrier height ($V_{\text{bias}} > \phi_B$), the tunneling mechanism is described by F-N tunneling⁷⁻¹⁰, and the F-N tunneling current density can be expressed by the formula of

$$J = \frac{q^3V_{\text{bias}}^2}{8\pi h q \phi_B d^2} \exp\left[\frac{-8\pi d(2qm^*)^{1/2}}{3hV_{\text{bias}}} \phi_B^{3/2}\right]. \quad (2)$$

To obtain the effective tunnel-barrier height information under different tunneling mechanisms, the above two formulas are linearized with the logarithmic scale, they can be written as:

$$\ln \frac{J}{V_{bias}^2} = \ln \left(\frac{3q^2(2m^*q\phi_B)^{\frac{1}{2}}}{2h^2d} \right) + \ln \left(\frac{1}{V_{bias}} \right) - \frac{4\pi d(2m^*q\phi_B)^{1/2}}{h} \text{ for direct tunneling. (3)}$$

$$\ln \frac{J}{V_{bias}^2} = \ln \frac{q^3}{8\pi h q \phi_B d^2} - \frac{8\pi d(2q m^* \phi_B^3)^{\frac{1}{2}}}{3h} \left(\frac{1}{V_{bias}} \right) \text{ for F-N tunneling. (4)}$$

Extracting from the J - V_{bias} data measured at 10 K (Figure 1b), the $\ln(J/V_{bias}^2)$ versus $1/V_{bias}$ curves for the different devices are shown in the **Figures S4a-f**, in which the red dotted line is fitted by the direct tunneling formula. Both device A and device B are dominated by direct tunneling in the range of 0-1.5 V bias. The devices C-F are dominated by direct tunneling under a small bias voltage, and gradually show F-N tunneling with the increase of bias voltage. Under a small bias voltage, the effective tunnel-barrier heights of the devices are obtained by fitting the Simmons' formula, which decreases slowly with the increase of the barrier thickness, and the effective tunnel-barrier height is ~ 0.9 eV (**Figure S4h**).

With the increase of the barrier thickness, the F-N tunneling become more and more obvious. For F-N tunneling, a plot of $\ln(J/V_{bias}^2)$ versus $1/V_{bias}$ should be linear. In addition, the slope of the F-N tunneling plots can be expressed as a function of the effective mass and the effective tunnel-barrier height:

$$\text{slope} = -6.83 \times 10^9 \times d \sqrt{\left(\frac{m^*}{m_0} \right) \phi_B^3}. \quad (5)$$

As shown in **Figure S4g**, the F-N tunneling plots of $\ln(J/V_{bias}^2)$ vs $1/V_{bias}$ with narrow abscissa value for the different devices, where the red dotted line is the curve fitted by the F-N tunneling formula. Therefore, the effective tunnel-barrier height of devices C, D, E, F and G can be estimated as 0.42, 0.45, 0.90, 0.88 and 0.86 eV, respectively. The effective tunnel-barrier height is also ~ 0.9 eV except for devices C and D, indicating transport mechanism is related to the barrier thickness⁸. In devices C and D, the direct tunneling current is large and cannot be ignored due to the their relatively small barrier thickness. Under a large bias voltage, device C and device D are jointly affected by direct tunneling and F-N tunneling. Thus, using the F-N tunneling formula to fit the effective tunnel-barrier heights of device C and device D will deviate. Since the bandgap of bulk GaSe is ~ 2 eV, the energy band diagram of the device under direct and F-N tunneling mechanisms are shown in the inset of **Figure S4h**. The band alignment with lager tunnel-barrier between the ferromagnetic metal and semiconductor is crucial for improving the TMR ratio of the devices; otherwise, the spin polarization would be diminished dramatically due to the conductivity mismatch^{11,12}.

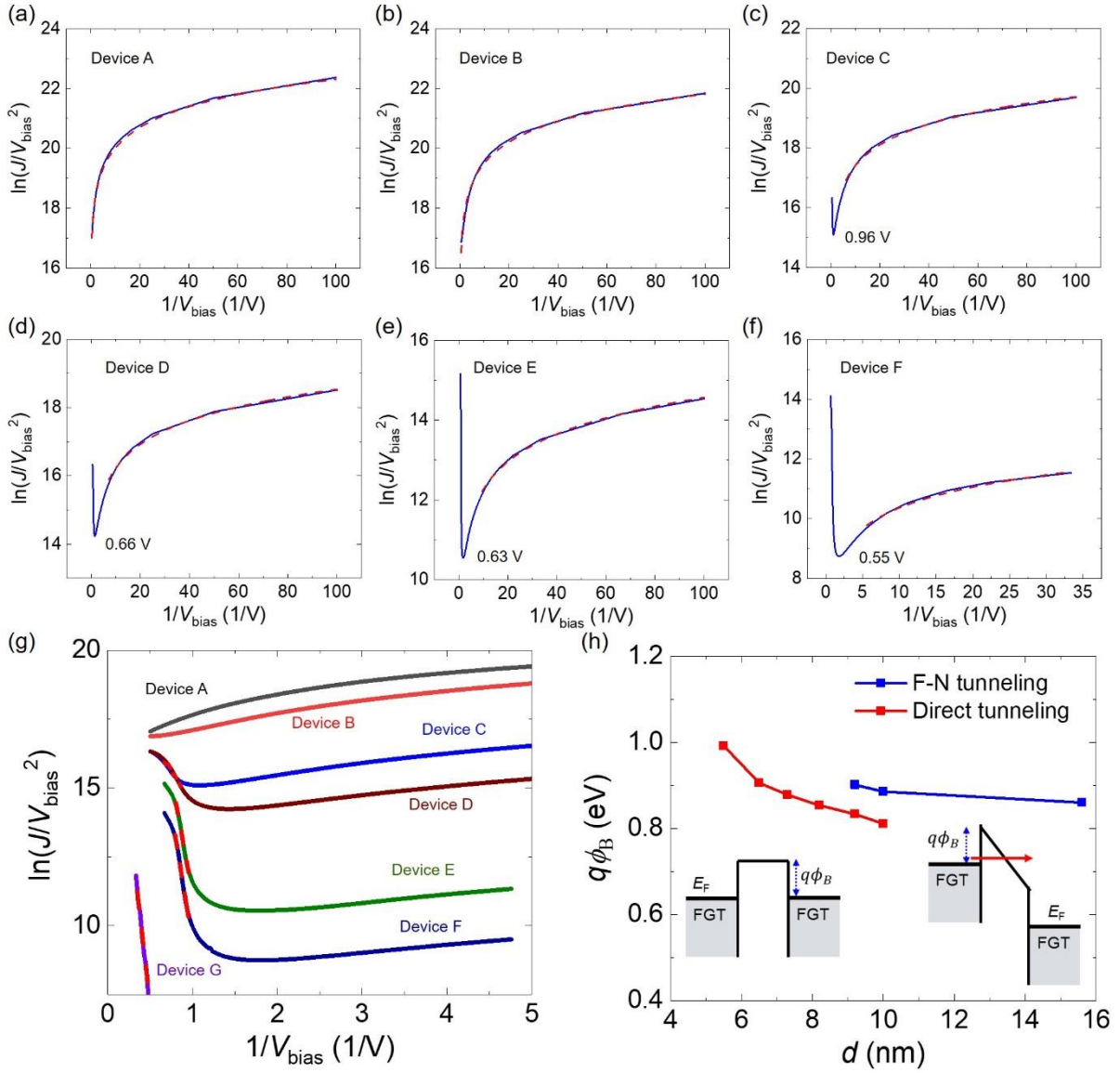


Figure S4. (a-f) The $\ln(J/V_{\text{bias}}^2)$ vs $1/V_{\text{bias}}$ curves for the different devices (blue line), and the red dotted line is fitted by the direct tunneling formula. (g) The F-N tunneling plots of $\ln(J/V_{\text{bias}}^2)$ vs $1/V_{\text{bias}}$ with narrow abscissa value for the different devices, where the red dotted line is the curve fitted by the F-N tunneling formula. (h) The effective tunnel-barrier height is extracted by fitting the direct tunneling formula and F-N tunneling formula respectively. The insets show the schematic diagram of direct tunneling under a small bias voltage and F-N tunneling mechanism under a large-bias voltage. The temperature is fixed at $T = 10$ K.

Supplementary Note 4. The measured TMR in the MTJ devices with different GaSe-layer thickness

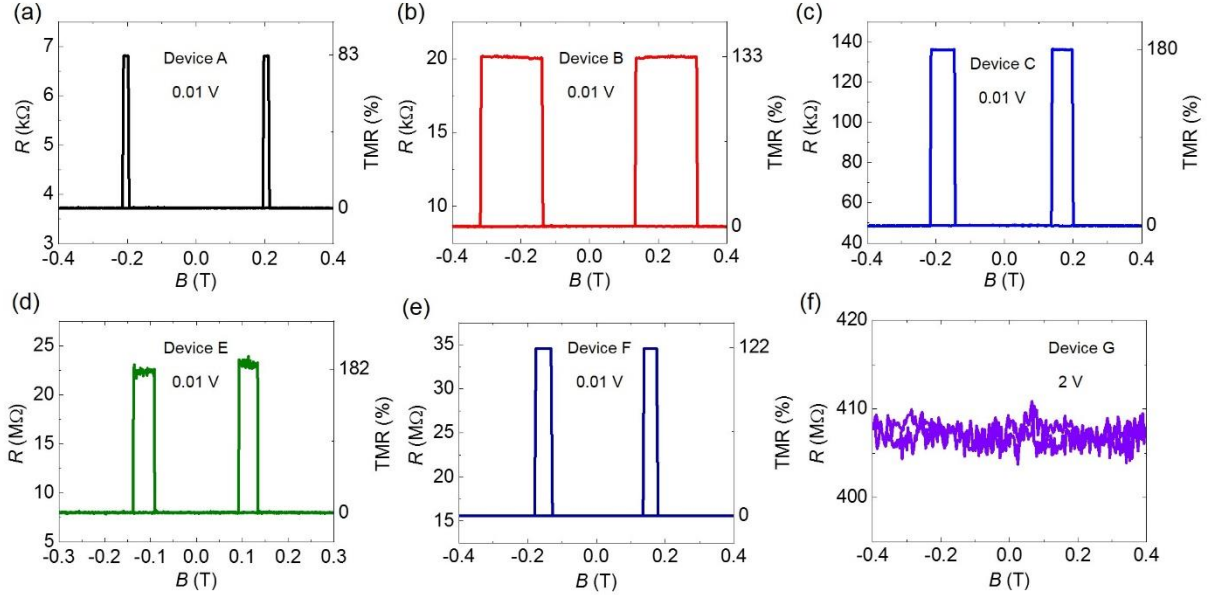


Figure S5. (a) R - B curves of device A at 10 mV. The R_P and R_{AP} of device A are 3.72 kΩ and 6.81 kΩ, with TMR ~83.1%. (b) R - B curves of device B at 10 mV. The R_P and R_{AP} of device B are 8.63 kΩ and 20.15 kΩ, with TMR ~133.5%. (c) R - B curves of device C at 10 mV. The R_P and R_{AP} of device C are 48.58 kΩ and 136.11 kΩ, with TMR ~180.2%. (d) R - B curves of device E at 10 mV. The R_P and R_{AP} of device E are 8.03 MΩ and 22.67 MΩ, with TMR ~182.3%. (e) R - B curves of device F at 10 mV. The R_P and R_{AP} of device F are 15.59 MΩ and 34.57 MΩ, with TMR ~121.7%. (f) R - B curves of device G under applied bias of 2 V. There is no TMR effect in device G. The temperature is fixed at 10 K.

Supplementary Note 5. The reproducibility of the barrier-thickness-dependent TMR

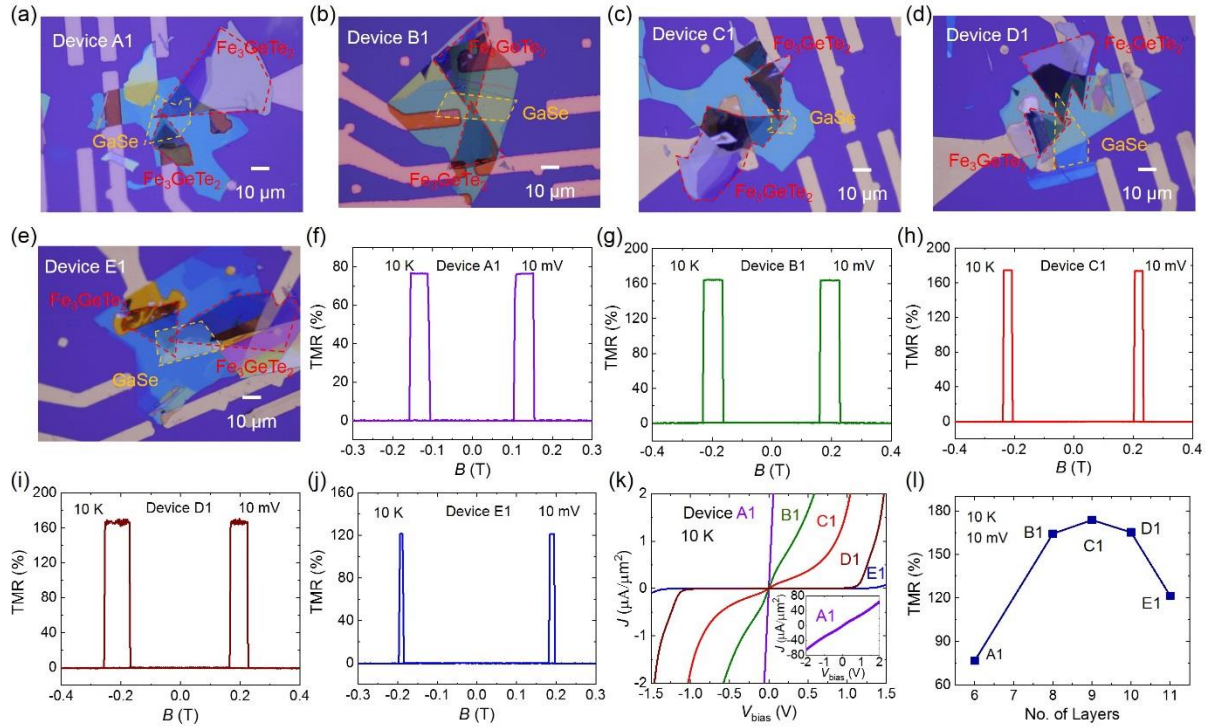


Figure S6 (a-e) The optical images of devices A1, B1, C1, D1, and E1, respectively. Scale bar: 10 μm. (f-j) The TMR- B curves of the devices measured at 10 K and 10 mV bias. (k) The J - V_{bias} curves at parallel state for the different devices. The inset shows the J - V_{bias} curve of the device A in a large bias range. (l) The extracted TMR value for the devices. The temperature is fixed at 10 K.

In order to confirm the behavior that TMR has a maximum value with varying the barrier layer thickness, we prepared another group of MTJ devices with different GaSe thickness ranging from 6-layers to 11-layers. Before preparing the ferromagnet/semiconductor heterojunction, we first determine the thickness of the used GaSe flakes through its optical contrast and AMF test results¹³. The optical images of devices A1, B1, C1, D1, and E1 are shown in **Figures S6a-e** respectively. The TMR- B curves of the devices measured at 10 K and 10 mV bias are shown in **Figures S6f-j**, respectively. The extracted TMR value is shown in **Figures S6l**. With the increased GaSe thickness, the TMR value first increases, then decreases, and the maximum value appears at 9-layer GaSe. As shown in **Figures S6k**, the J - V_{bias} curves at parallel state for different devices is still nonlinear and dispersed, indicating the transport is determined by the tunneling mechanism. In summary, the reproducibility of the barrier-thickness-dependent TMR characteristics has been proved.

Supplementary Note 6. The bias-dependent TMR in the MTJ devices

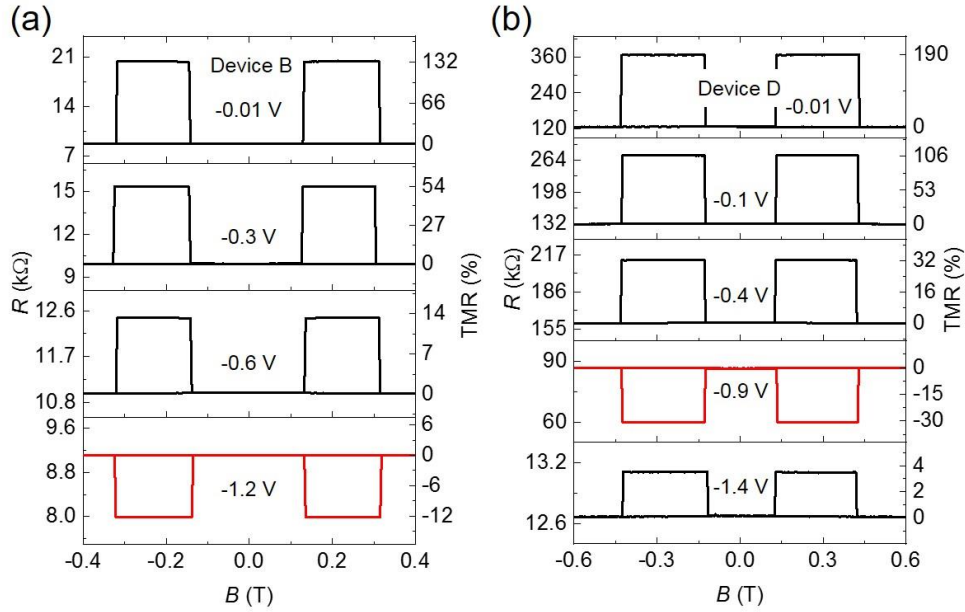


Figure S7. R - B curves and the corresponding TMR at various negative bias for devices B with 6.5-nm-thick GaSe (a) and D with 8.2-nm-thick GaSe (b) at 10 K.

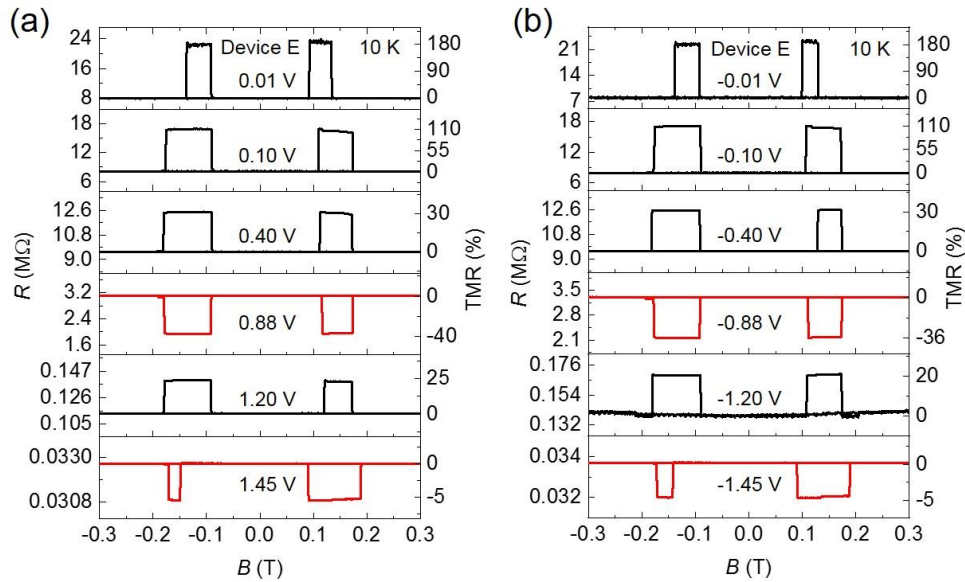


Figure S8. R - B curves and the corresponding TMR of device E with 9.2-nm-thick GaSe-layer at various positive bias (a) and negative bias (b). The temperature is fixed at 10 K.

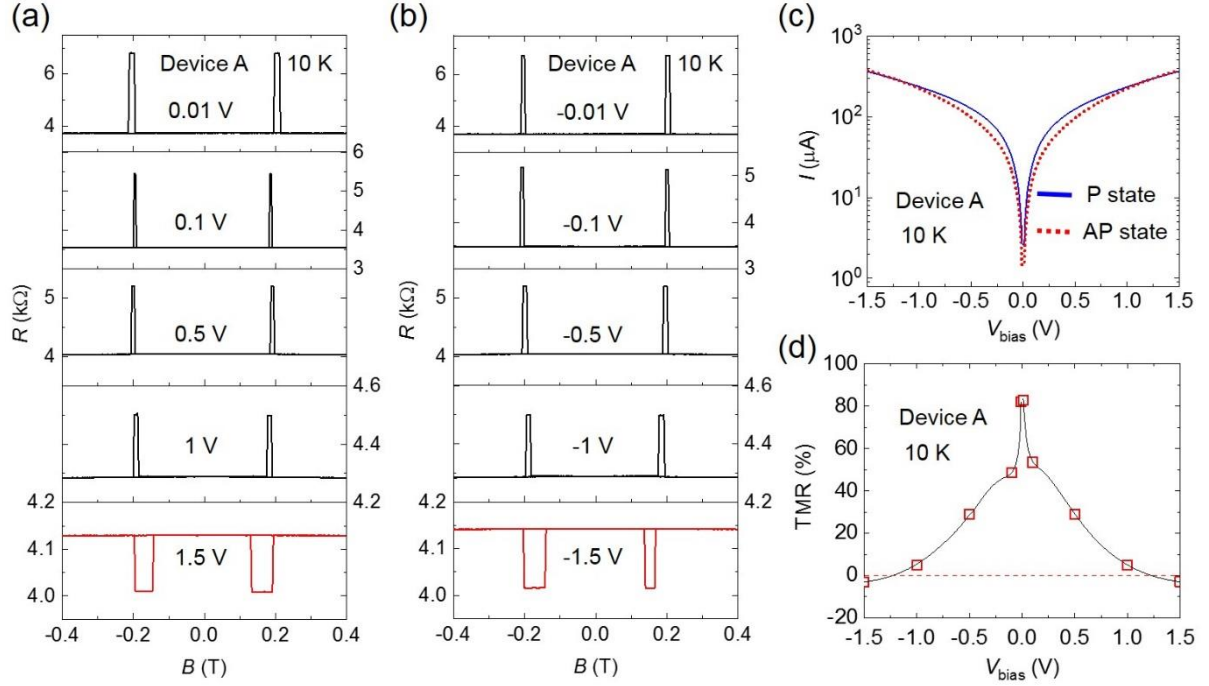


Figure S9. R - B curves of device A with 5.5-nm-thick GaSe-layer at various positive bias (a) and negative bias (b). (c) I - V_{bias} curves of device A measured in parallel and antiparallel magnetic configurations, respectively. (d) TMR of device A as a function of V_{bias} , which decreases with the increase of bias and becomes negative when the bias exceeds 1.20 V. The temperature is fixed at 10 K.

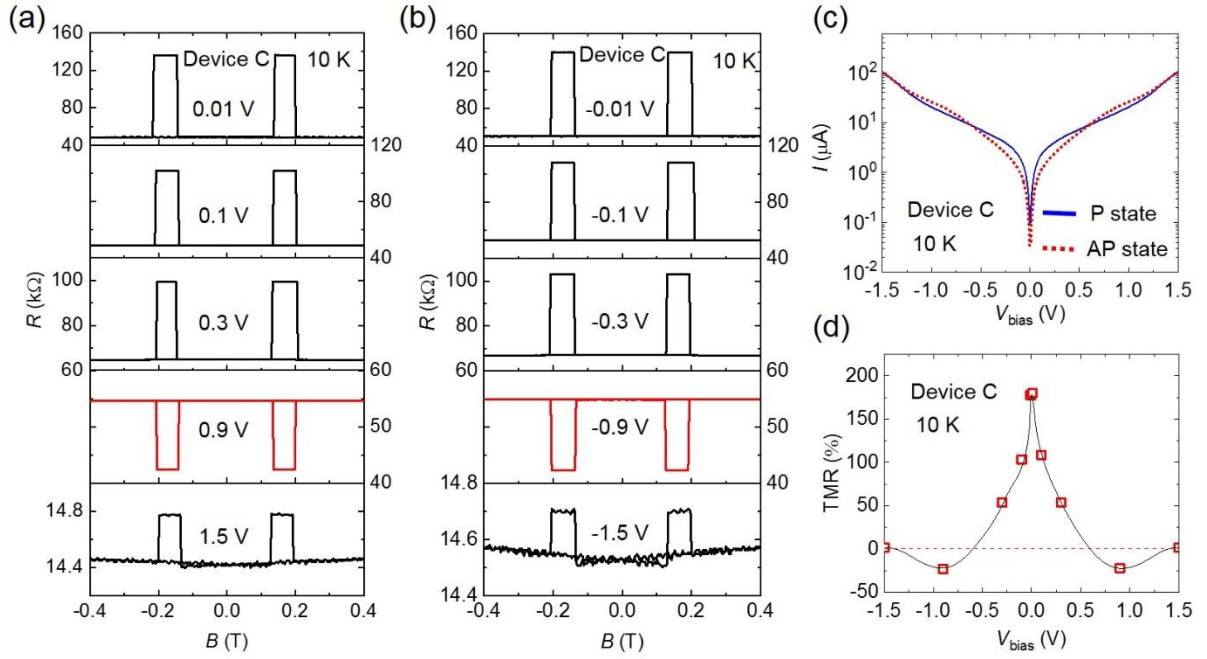


Figure S10. R - B curves of device C with 7.3-nm-thick GaSe-layer at various positive bias (a) and negative bias (b). (c) I - V_{bias} curves of device C measured in parallel and antiparallel magnetic configurations, respectively. (d) TMR of device C as a function of V_{bias} , which decreases with the increase of bias and becomes negative when the bias exceeds 0.60 V and then back to positive value when the bias exceeds 1.43 V. The temperature is fixed at 10 K.

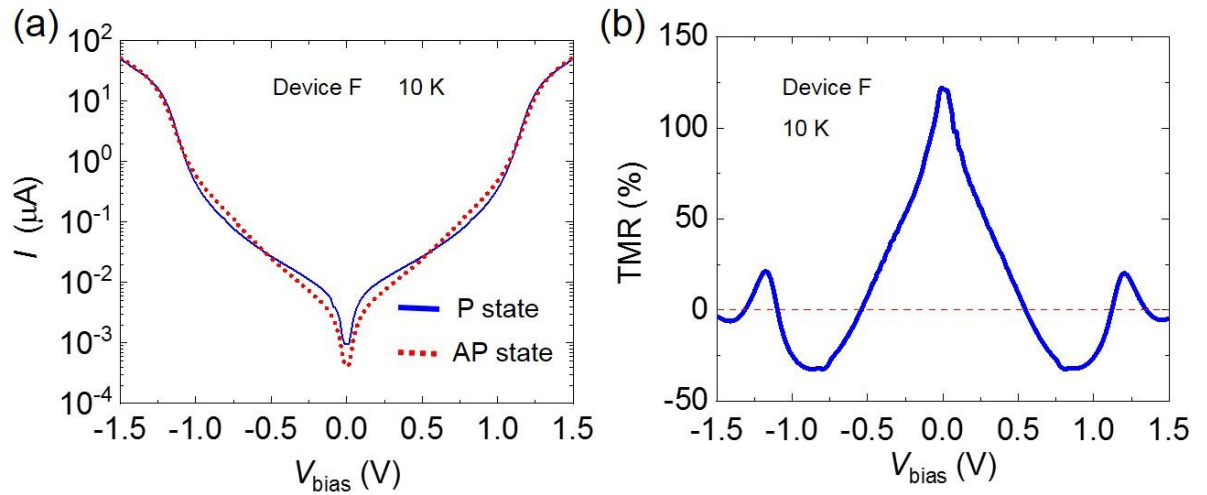


Figure S11. (a) I - V_{bias} curves of device F with 10-nm-thick GaSe-layer in parallel and antiparallel states, respectively. (b) TMR versus V_{bias} of device D. The temperature is $T = 10$ K.

Supplementary Note 7. The spin-resolved DOS simulated by DFT

Ab Initio Calculations: Our first-principles calculations are performed by the density functional theory (DFT) using the Vienna *ab initio* Simulation Package (VASP) code^{14,15}. The exchange and correlation functionals are implemented by the generalized gradient approximation (GGA) of the Perdew-Burke-Ernzerhof functional¹⁶. A 2D FGT/GaSe/FGT van der Waals (vdW) heterostructure is constructed by sandwiching a 6-layer-GaSe between two 3-layer-FGTs. The lattice constant for pure FGT and GaSe is 3.99 and 3.81 Å, respectively. To obtain the correct spin polarization of FGT, the lattice constant of FGT/GaSe/FGT heterostructure is chosen as for the lattice of FGT, 3.99 Å, with a lattice mismatch of 4.7% compared to that of GaSe. Three different configurations between FGT and GaSe are considered: (i) Ga and Se atoms in GaSe-layer atop on Fe and Te atoms in FGT layer, respectively, (ii) Ga and Se atoms in GaSe-layer atop on Ge and Fe atoms in FGT layer, respectively, and (iii) Ga and Se atoms in GaSe-layer atop on Te and Ge atoms in FGT layer, respectively. The corresponding energy of these arrangements is -53.369, -53.440, and -53.448 eV, respectively. We find that the most stable configuration is (iii), as shown in **Figure S12**. The thickness of vacuum layer about 15 Å is set to avoid interactions between adjacent layers. Structures are fully relaxed until the force converged on each atom to less than 10^{-2} eV/Å. In all calculations, the plane-wave cutoff energy is set to 350 eV; Brillouin zone is sampled using Γ -centered $15 \times 15 \times 1$ Monkhorst-pack k mesh¹⁷; and the vdW interaction are corrected by DFT-D3 Grimme method¹⁸.

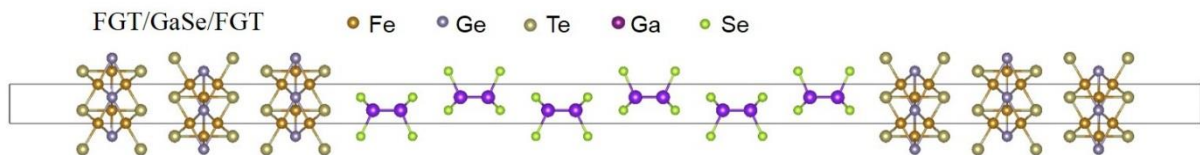


Figure S12. A side view of 3-layer-FGT/6-layer/3-layer-FGT atomic structure used for the calculations.

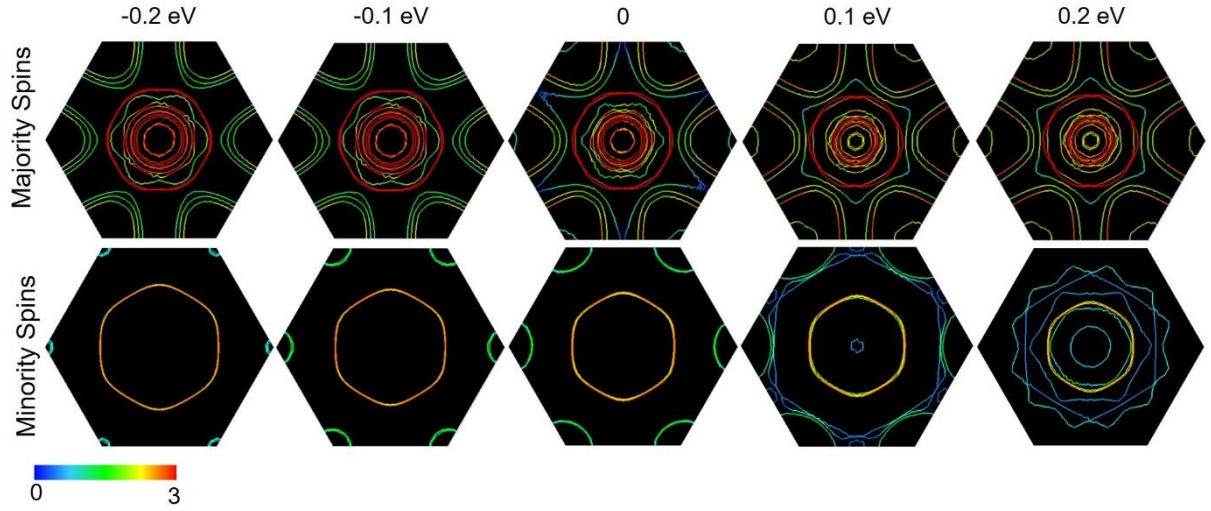


Figure S13. Majority and minority spins Fermi surfaces of three-layer Fe_3GeTe_2 at different energies ranging from $E_F-0.2$ to $E_F+0.2$ eV, where “0” indicates the Fermi energy. Colors indicate the Fermi surfaces belonging to different bands.

Figure S13 shows the majority and minority spins Fermi surfaces of three-layer Fe_3GeTe_2 at different energies ranging from $E_F-0.2$ to $E_F+0.2$ eV, where “0” indicates the Fermi energy. Majority spins have multiple bands at the Fermi energy covering the large portion of the 2D Brillouin zone (BZ), the minority spins have only a few states available, resulting in a large area of the 2D BZ with no overlap. Based on the Julliere model, this will give a large positive magnetoresistance. When voltage is applied to the MTJ device, the energy surface of one of the Fe_3GeTe_2 electrodes will move relatively to the other electrode, which makes the majority spins (or minority spins) of one electrode overlap with the minority spins (or majority spins) of the other electrode. As shown in **Figure S13**, in case of $E_F+0.2$ eV, some bands of majority and minority spins overlaps. As a result, the negative magnetoresistance will appear.

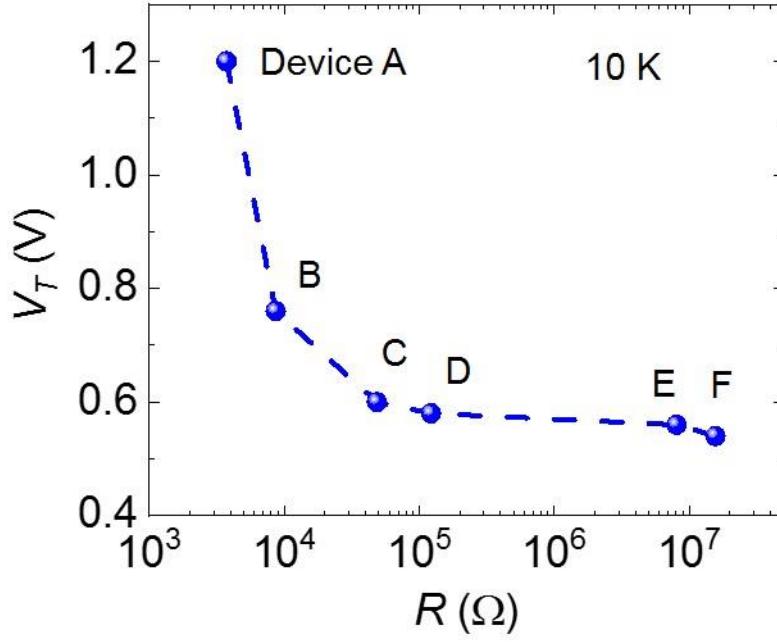


Figure S14. Bias voltage of the first TMR sign reversal at 10 K as a function of device resistance.

To investigate the relationship between thickness-dependent TMR sign reversal and device resistance, we extracted the bias voltage values of the first TMR sign reversal at 10 K as a function of device resistance, as shown in **Figure S14**. As the device resistance increases exponentially, the bias voltage values first rapidly decrease and then gradually stabilize. For devices A and B with lower resistance, the voltage dropped on tunnel junction is significantly smaller than the voltage applied, thus a larger bias voltage is required to reach the transition point of the first TMR sign reversal. With increasing the junction resistance, the voltage-divider effect becomes less important. This is why the difference of the voltage for the first TMR sign reversal between device D and E is very small.

Supplementary Note 8. The temperature-dependent TMR in the MTJ devices

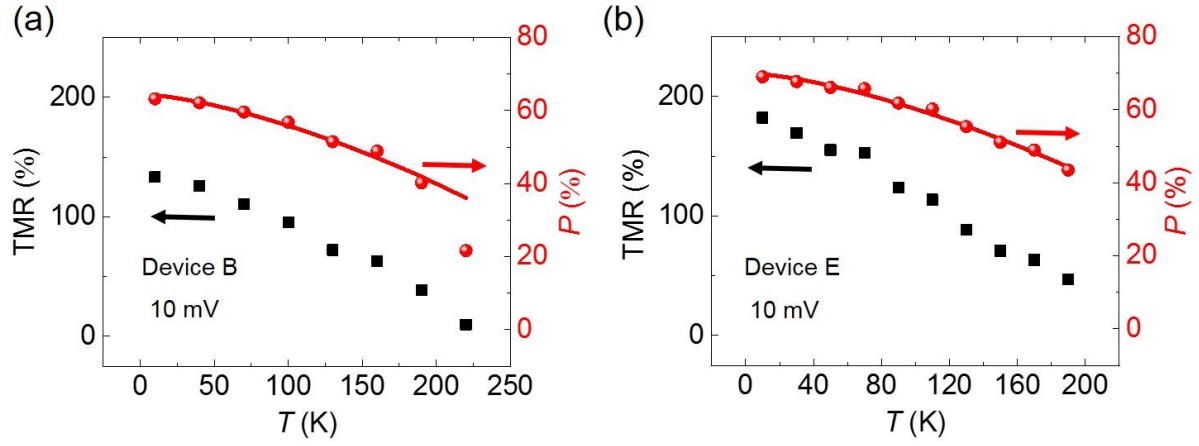


Figure S15. (a) The extracted TMR and the corresponding P of device B as a function of temperature at bias-voltage of 10 mV. The fitting value of α is $1.34 \times 10^{-4} \text{ K}^{-3/2}$. (b) The extracted TMR and the corresponding P of device E as a function of temperature at bias-voltage of 10 mV. The fitting value of α is $1.40 \times 10^{-4} \text{ K}^{-3/2}$.

References for Supplementary Information

1. Li, X. et al. Controlled Vapor Phase Growth of Single Crystalline, Two-Dimensional GaSe Crystals with High Photoresponse. *Sci. Rep.* **4**, 5497 (2014).
2. Cao, Y. et al. Strong enhancement of photoresponsivity with shrinking the electrodes spacing in few layer GaSe photodetectors. *Sci. Rep.* **5**, 8130 (2015).
3. Molas, M. R. et al. Raman spectroscopy of GaSe and InSe post-transition metal chalcogenides layers. *Faraday Discuss.* **227**, 163-170 (2021).
4. Hu, P. et al. Synthesis of Few-Layer GaSe Nanosheets for High Performance Photodetectors. *ACS Nano* **6**, 5988-5994 (2012).
5. Simmons, J. G. Generalized Formula for the Electric Tunnel Effect between Similar Electrodes Separated by a Thin Insulating Film. *J. Appl. Phys.* **34**, 1793-1803 (1963).
6. Le Toullec, R., Piccioli, N. & Chervin, J. C. Optical properties of the band-edge exciton in GaSe crystals at 10 K. *Phys. Rev. B* **22**, 6162 (1980).
7. Yıldırım, M. & Öksüzöğlu, R. M. Effect of bias voltage on tunneling mechanism in Co₄₀Fe₄₀B₂₀/MgO/Co₄₀Fe₄₀B₂₀ pseudo-spin valve. *J. Magn. Magn. Mater.* **379**, 280-287 (2015).
8. Chiu, F.-C. A Review on Conduction Mechanisms in Dielectric Films. *Adv. Mater. Sci. Eng.* **2014**, 578168 (2014).
9. Kim, S. et al. Highly Efficient Experimental Approach to Evaluate Metal to 2D Semiconductor Interfaces in Vertical Diodes with Asymmetric Metal Contacts. *ACS Appl. Mater. Interfaces* **13**, 27705-27712 (2021).
10. Morgan, C. et al. Impact of Tunnel-Barrier Strength on Magnetoresistance in Carbon Nanotubes. *Phys. Rev. Appl.* **5**, 054010 (2016).
11. Rashba, E. I. Theory of electrical spin injection: Tunnel contacts as a solution of the conductivity mismatch problem. *Phys. Rev. B* **62**, R16267(R) (2000).
12. Dolui, K., Narayan, A., Rungger, I. & Sanvito, S. Efficient spin injection and giant magnetoresistance in Fe/MoS₂/Fe junctions. *Phys. Rev. B* **90**, 041401(R) (2014).
13. Bing, D. et al. Optical contrast for identifying the thickness of two-dimensional materials. *Opt. Commun.* **406**, 128-138 (2018).
14. Kresse, G. & Hafner, J. Ab initio molecular-dynamics simulation of the liquid-metal--amorphous-semiconductor transition in germanium. *Phys. Rev. B* **49**, 14251 (1994).
15. Kresse, G. & Joubert, D. From ultrasoft pseudopotentials to the projector augmented-wave method. *Phys. Rev. B* **59**, 1758 (1999).
16. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **77**, 3865 (1996).

17. Monkhorst, H. J. & Pack, J. D. Special points for Brillouin-zone integrations. *Phys. Rev. B* **13**, 5188 (1976).
18. Grimme, S., Antony, J., Ehrlich, S. & Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu. *J. Chem. Phys.* **132**, 154104 (2010).